

Why Learn AI?

Pretend you're a scribe in the year 1500. All day, you hand-copy the king's proclamations and send them out across the kingdom. Good work. Steady job. Then one morning, you hear that a guy in Mainz built a machine that prints pages a thousand times faster than you can write them. Two choices: pretend it isn't happening, or learn the new machine well enough to run it.

AI is the press. Run it, or someone else will.



AI IS EVERYWHERE

AI isn't something you go visit. It's already in the apps on your phone, the search results you read, and the tools your first job will hand you on day one. It feels like AI went from sci-fi to normal in about two years. **Most of what it does today, it will do better tomorrow.**

AI is already in the apps you use every day



Recommends

Predicts what you might like next.

Spotify, Netflix, TikTok



Navigation

Predicts traffic and arrival time.

Google Maps, Waze



Face recognition

Checks whether a face matches you.

Phone unlock, photo tagging



Voice assistants

Turns your speech into words.

Siri, Alexa, Hey Google



Chatbots

Carries on a conversation with you.

ChatGPT, Claude, Gemini

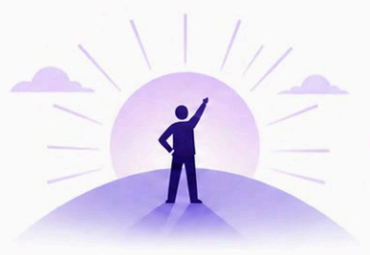
AND IT'S EARLY

Why you'll thrive in the AI future

1

THIS IS YOUR TIME

Nobody has a twenty-year head start. That almost never happens with something this big. You're showing up right as it lands.



2

YOU'LL MOVE FASTER

AI collapses years of paying dues, learning the trade, and climbing the ladder. What took a decade is within reach now.



3

NOTHING TO UNLEARN

Others must undo the workflow that made them fast. You skip all of that and learn the new way from the start.



THIS HAS HAPPENED BEFORE

Before personal computers, becoming a designer meant years at a drafting table learning the craft by hand. Then desktop publishing showed up, and a teenager with a Mac could turn out professional work in an afternoon. The tool didn't replace skill, it shortened the distance between wanting to do the work and actually doing it. And it's the rule, not the exception: the steam engine did it for physical labor, electricity for the factory, the internet for information. **AI is that, for almost everything.**

And this isn't just tech industry hype. In July 2025, the White House released an official national strategy document, *Winning the Race: America's AI Action Plan*. Here's how it describes what AI makes possible:

“

An industrial revolution, an information revolution, and a renaissance —all at once. This is the potential that AI presents.

— **WINNING THE RACE** · The White House, July 2025

AI is today's printing press.

Learn to run it.

What Is AI?

You hear about AI constantly. So what is it, really? The useful starting point: AI is software built to do things that used to take a human brain. And it comes in two kinds.

Types of AI (and you already use both)

 **SORTS & RECOMMENDS**

Recommendation AI

Chooses from what already exists



Ranks the options and picks the top one

WHERE YOU MEET IT

Next show on Netflix

Next song on Spotify

Fastest route on Maps

Never makes anything new.

 **CREATES NEW CONTENT**

Generative AI

Makes something that didn't exist

Twin boys in high school, who love the Dallas Stars, are building an AI course at a warm kitchen table.

This prompt made the picture on the welcome page.

WHAT IT CAN DO

Writes the email

Drafts the essay

Makes the image

Codes the website

Creates the song

Builds the video

When people picture AI, they usually mean this, and it's what this course is about.

Here's what that difference looks like on one task.

THE TASK

“What movie should I watch tonight?”



Recommendation AI

Netflix-style recommendation

Midnight Harbor

seen

Echoes of the Tide

TOP PICK

The Glass Orchard

Northern Lights

It just moves to the next title on the list. It has no idea why you skipped the first.



Generative AI

Writes a reply, in a conversation

How about Midnight Harbor? It's the slow-burn thriller most people rate highest right now, rainy and tense the whole way through.

Composes a reply, and you can talk back.

 I already saw that at a friend's.

Ah, then skip it. Try Paper Cranes instead, it keeps that same rainy, tense mood but it's a quiet little mystery you've almost certainly missed. Want something lighter?

Recommendation AI just slid to the next title on its list. Generative AI took your situation, that you'd already seen it, and composed something new from it. And it's still in the conversation.



THE NAME GAME

You'll hear a pile of names: AI, generative AI, ChatGPT, LLM. Don't let the terminology trip you up. For the rest of this course, we'll usually say AI, and we mean the generative kind that you experience with ChatGPT, Claude, and Gemini.

There's a name for these generative AI tools: LLM, short for **Large Language Model**. Think of it like this. ChatGPT is the app you use, and the LLM is the engine under the hood doing the work.

What's an LLM?

L

Large

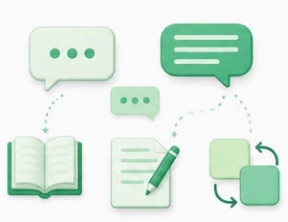
Trained on huge amounts of text and code.



L

Language

Reads, writes, summarizes, translates, and explains.



M

Model

Predicts likely output from learned patterns.



ChatGPT is the app. The LLM is the engine.

Two kinds. One picks, one makes.

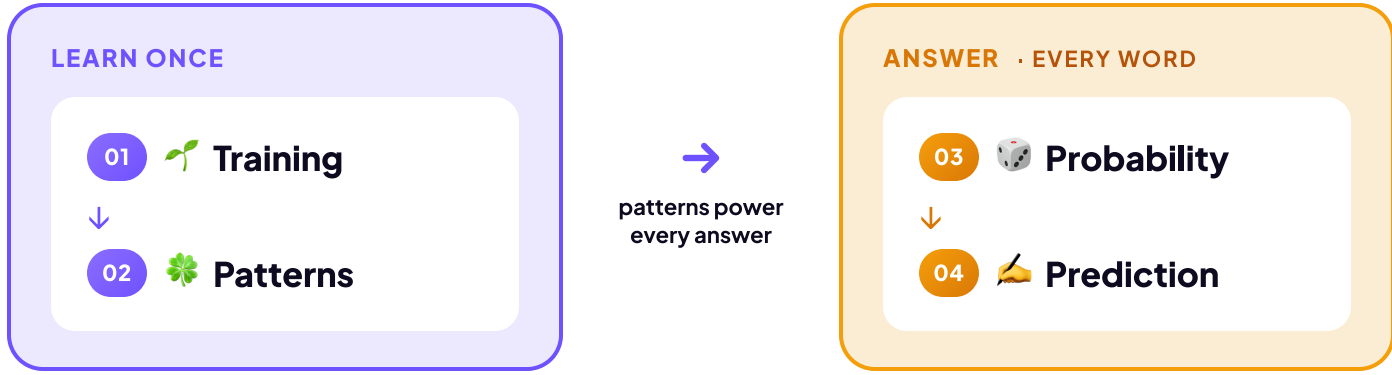
This course is about the one that makes.

How an LLM Works

You've met the two kinds of AI and named the generative one: a Large Language Model, or LLM. So how does an LLM actually turn your words into an answer?

When ChatGPT or Claude write a sentence, they're running math to predict the likely next words. They aren't looking up what your words mean; they're working out which words tend to follow which.

How do they do it? In two phases. First the model learns, once, by soaking up patterns from mountains of text. Then, every time you chat, it uses those patterns to build your answer one word at a time.



Four ideas carry it. Below, we follow one example, peanut butter, through all four.

01 Training LEARN ONCE

The model **teaches itself**: guess the next word, check, and nudge its numbers toward the right word.



Repeat that loop billions of times, and that's training.

02 Patterns LEARN ONCE

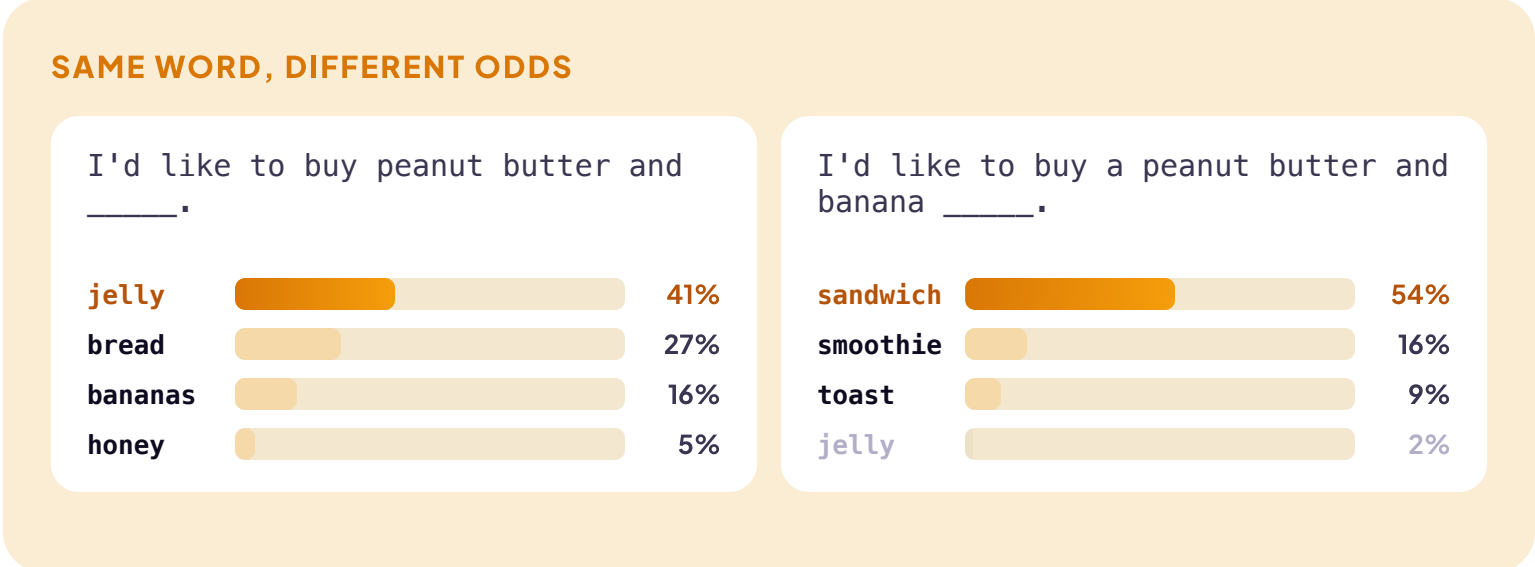
So what is it actually learning? **Patterns**. Here's one you picked up as a child. Which word comes next?



AI didn't memorize these. It absorbed the pattern by running through billions of examples.

03 Probability EVERY WORD

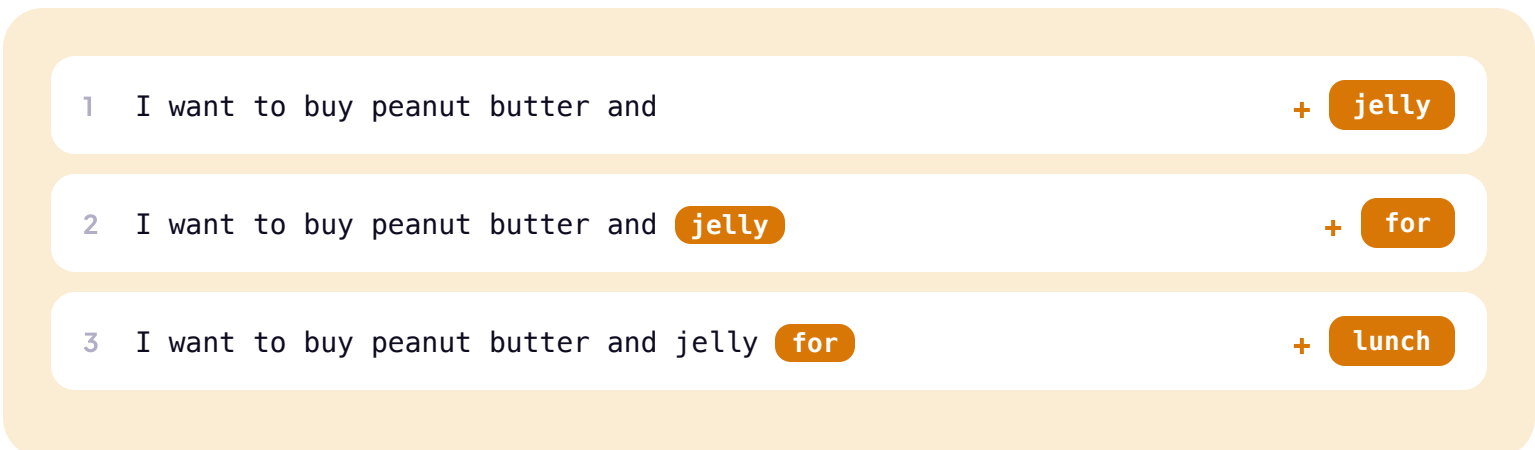
AI doesn't make one guess. It scores **every** possible next word: a ranked list with a probability on each, and those numbers shift with the surrounding text.



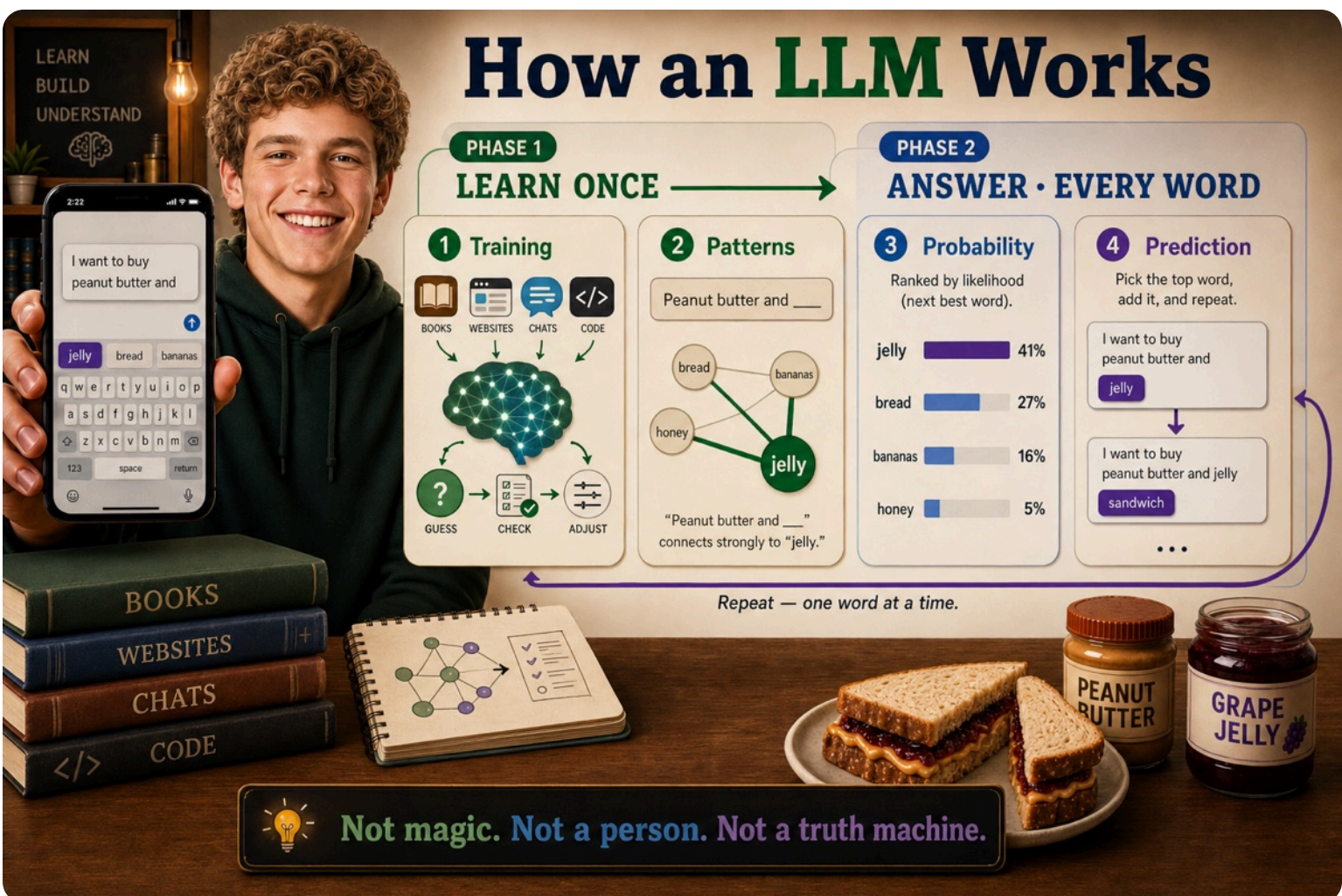
Add the word banana, and jelly's probability drops from 41% to 2%. The word banana changed everything.

04 Prediction EVERY WORD

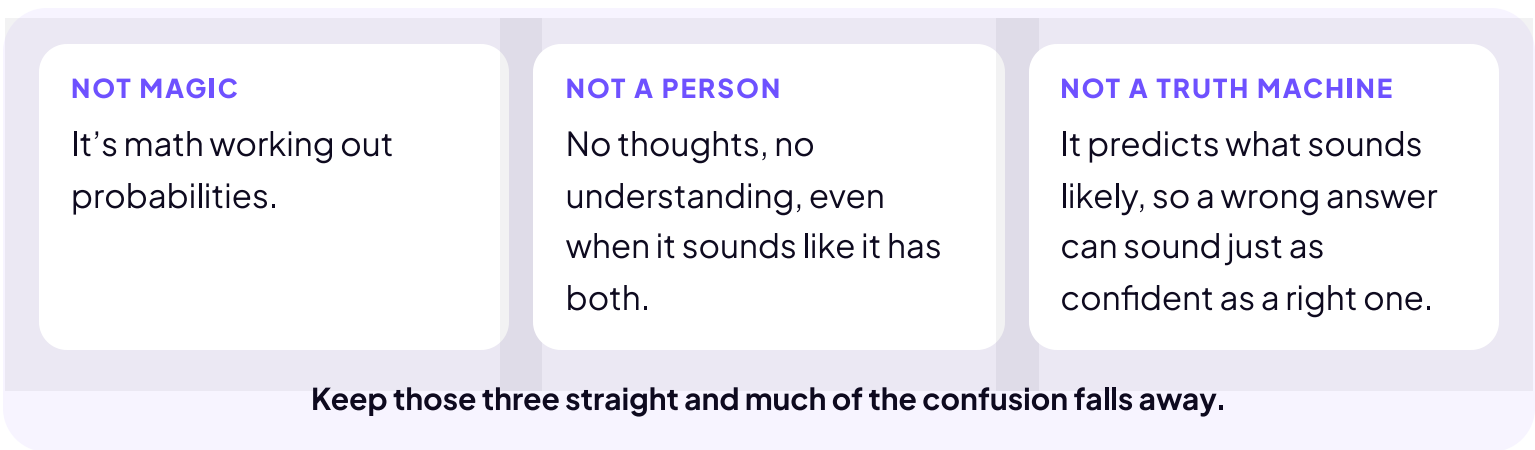
Probability handled one word. But your answer is hundreds of words long, so the model just repeats the move. Your phone does this when you write a text: it suggests a word, you tap it, it suggests the next. AI works the same way, with no fixed plan for where the sentence will end up. One word, look again, the next.



A full paragraph runs this loop many times, fast enough to look like thought.



That clears up three big myths:



Keep those three straight and much of the confusion falls away.

Not magic. Not a person. Not a truth machine.

It's math working out probabilities.

Does AI Think?

You type a question and AI types back like a person. It explains, it jokes, it says sorry when it slips up. After a while it’s hard not to feel like someone is in there. So here’s the big question: **Does AI actually think?** Is it alive? Is it smart? Does it feel anything?

Short answer: no, not the way you do.

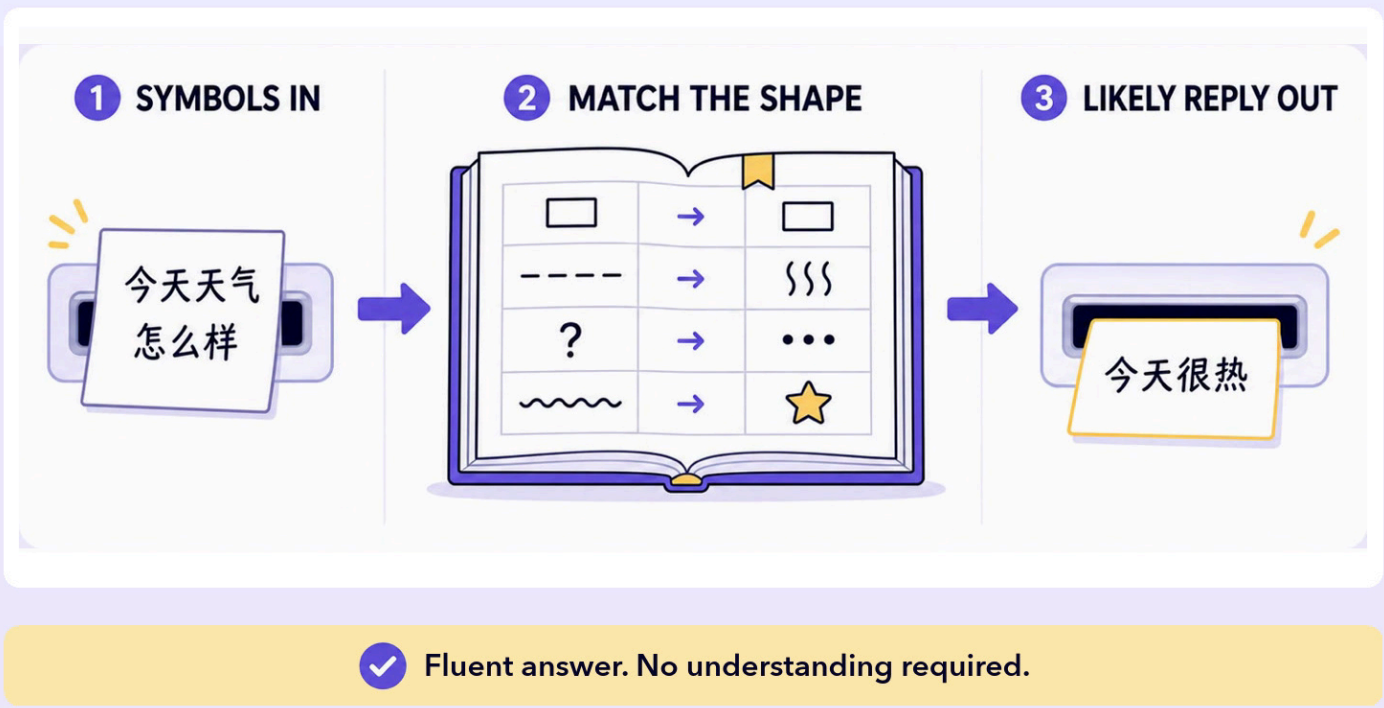
IT’S IN THE NAME

We call it Artificial Intelligence, not Artificial Thinking. That was picked on purpose. Something can act intelligent, finish your sentence, explain a poem, without understanding a single thing it just said.

There’s a famous way to picture this, sometimes called the Chinese Room. Imagine someone locked in a room who doesn’t know a word of Chinese. Notes in Chinese come in under the door. The person can’t read any of it, but they have a giant rulebook telling them which symbols to send back. So they match symbols and slide back an answer that looks perfect to the people outside.



Matching the pattern is not understanding it



That’s AI. It arranges symbols brilliantly and understands none of them. You get a perfect-looking answer with no real understanding behind it.

When you think vs. what AI does

 A HUMAN MIND	 A PATTERN MACHINE
When you think	What AI does instead
Understand what words really mean	Matches patterns in mountains of text
Draw on real experience	Only reads about the world
Choose words to make a point	Predicts the next likely word
Feel when something’s beautiful	Echoes what others call beautiful
Know when you’re unsure	Can’t tell when it’s making things up

None of this means AI is dumb or useless. A pattern machine this good is genuinely powerful. It just means it’s not thinking.

A pattern machine, not a thinker.

Once you see that, its weird moments make sense.

What You Can Control

You’re hearing it constantly: AI is taking jobs. That headline used to be a prediction. Now it’s news.

And jobs are just the loudest headline. AI is also reshaping who holds power, what it costs the planet, and what you can trust online. Which way it all goes, nobody knows yet, including the people making the predictions. And almost none of it is in your hands.

So here’s the question worth asking: what’s in your hands, and what isn’t?

What’s actually in your hands?

OUT OF YOUR HANDS



Headlines and hype

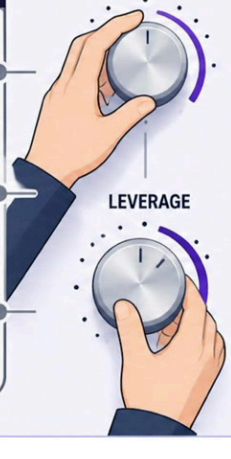


How fast AI improves



How persuasive AI gets

VOLUME



LEVERAGE

IN YOUR HANDS



How much the hype gets to you



How well you master AI



Whether you do your own thinking

 The left side is loud. The right side is leverage.

Most of the noise around AI is the left column. Most of the leverage in your life is the right column.

SO WHAT DO YOU DO?

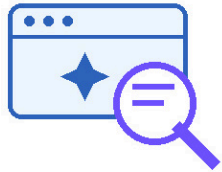
That right column isn’t a feeling, it’s a to-do list. Three moves worth your energy:

Three moves worth your energy

1

Get genuinely good with AI

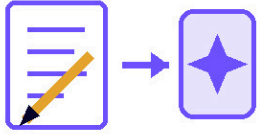
Pick one tool and go deep. Learn its strengths, limits, and how to push it.



2

Think first, then bring AI in

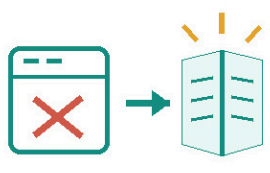
Form your own take before you ask. Use AI to sharpen it, not skip the thinking.



3

Close the tab and go learn

Trade the doomscrolling hour for getting better at something real.



Controlling your skill and judgment isn’t a consolation prize for the stuff you can’t change. It’s the one lever that actually tilts your odds, even when the big forces aren’t yours.

Your skill and judgment aren’t a consolation prize.

They’re the mechanical lever that tilts the odds in your favor.

Does School Matter?

A lot of people your age are quietly asking the same thing. If AI can answer anything, do you even need school anymore? Why sit through years of classes when the tool spits out the answer in seconds?

Think about it this way.

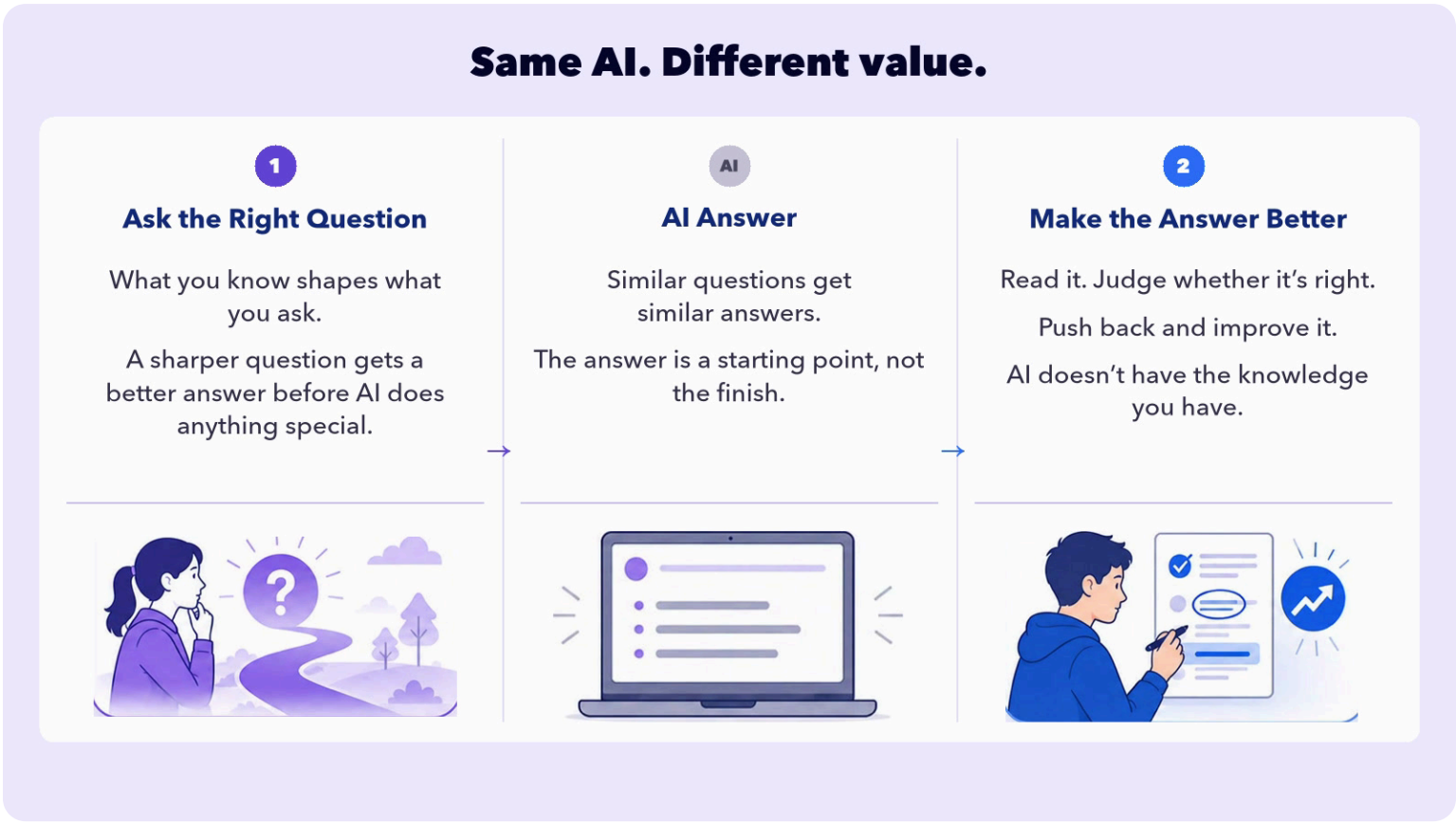
Fast forward a few years. You’ve landed your dream job, and AI sits right next to you all day, cranking out drafts, code, and plans. Your coworker at the next desk is doing the same. You each have the same job, using the same AI.

Here’s the catch: ask AI a similar question, and it hands back a similar answer, to you, your coworker, and everyone else. So what makes you more valuable? It’s not the answer itself. **It’s what you already know, which shapes the question you ask and what you do with the answer.**

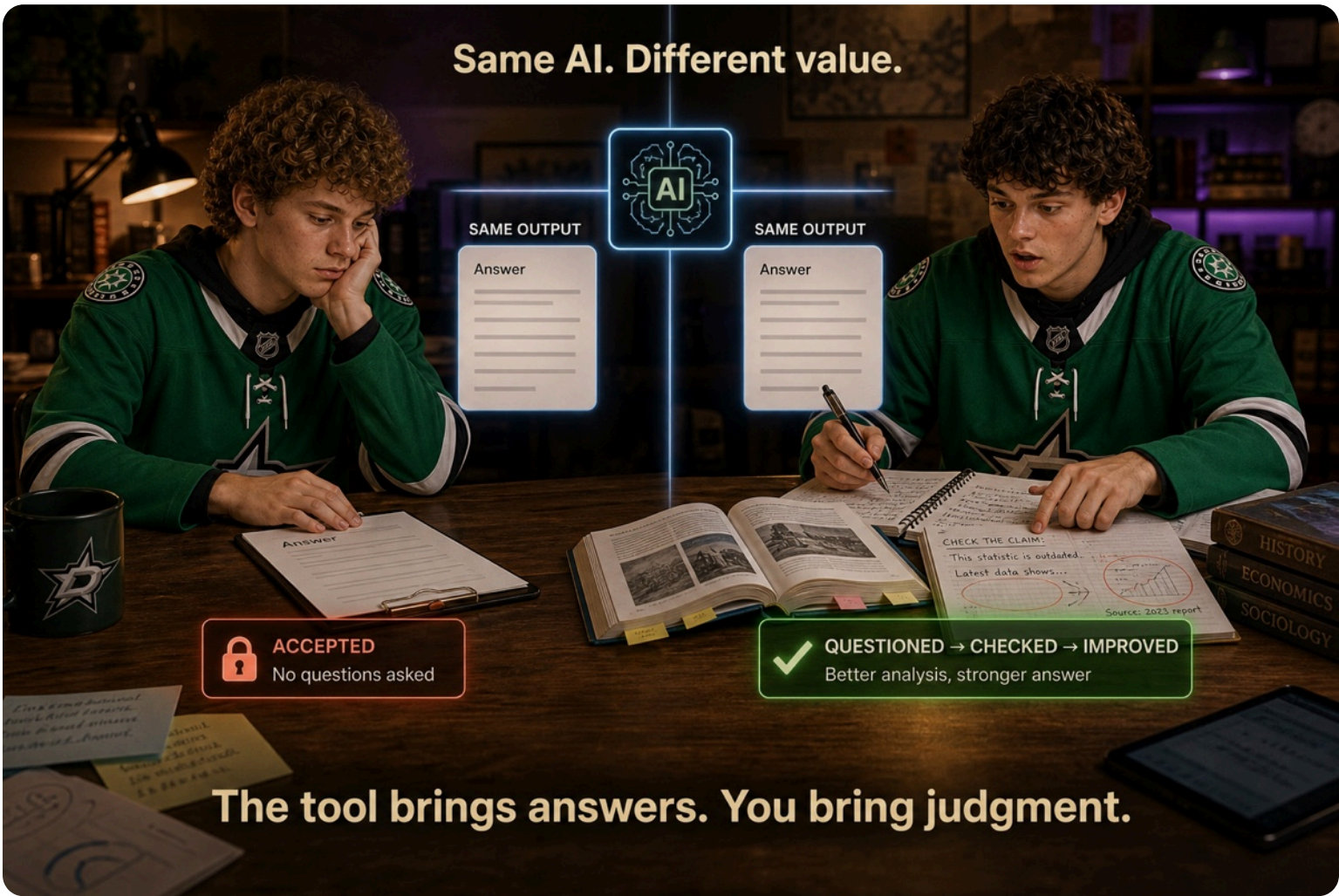
WHAT MOST PEOPLE DO NOT UNDERSTAND

In the AI era, learning is more important than ever. This seems counterintuitive, but it’s true. This inspired the name of this course, **Be Smarter Than the Tool**. Whatever college and career you choose, the more you learn, the more successful you’ll be.

For almost any career, you’ll have AI as your partner. So here’s what it looks like.



Those two things are your domain. This is where you need to live. Double down on building your knowledge and skills. And one day you’ll be the CEO at Google.



The tool brings answers. You bring judgment.

It starts today, not when you land the job.

Learn with AI

Here's the good news: when it comes to studying, AI might be the most powerful tool you'll ever get your hands on. Used the right way, it doesn't just help you finish the work, it makes you genuinely sharper at the subject.

It's the most patient tutor you'll ever meet. It'll quiz you at 1am, never sigh when you ask again, and meet you exactly where you're stuck.

The first move is choosing which kind of AI to aim at the job. There's a simple question to guide your choice: do you want to learn from materials you already have, or learn something new?

Which study tool for the job?

	YOUR MATERIALS	SOMETHING NEW
	Source-Grounded Tutor GEMINI NOTEBOOK	General Tutor CHATGPT · CLAUDE · GEMINI
WHAT IT IS	Learns only from the materials you provide.	Explains and quizzes from broad training.
BEST FOR	Exam prep from notes, guides, slides, and class links.	New explanations and extra practice.
THE CATCH	Missing source means missing knowledge.	Can make things up or differ from your class.

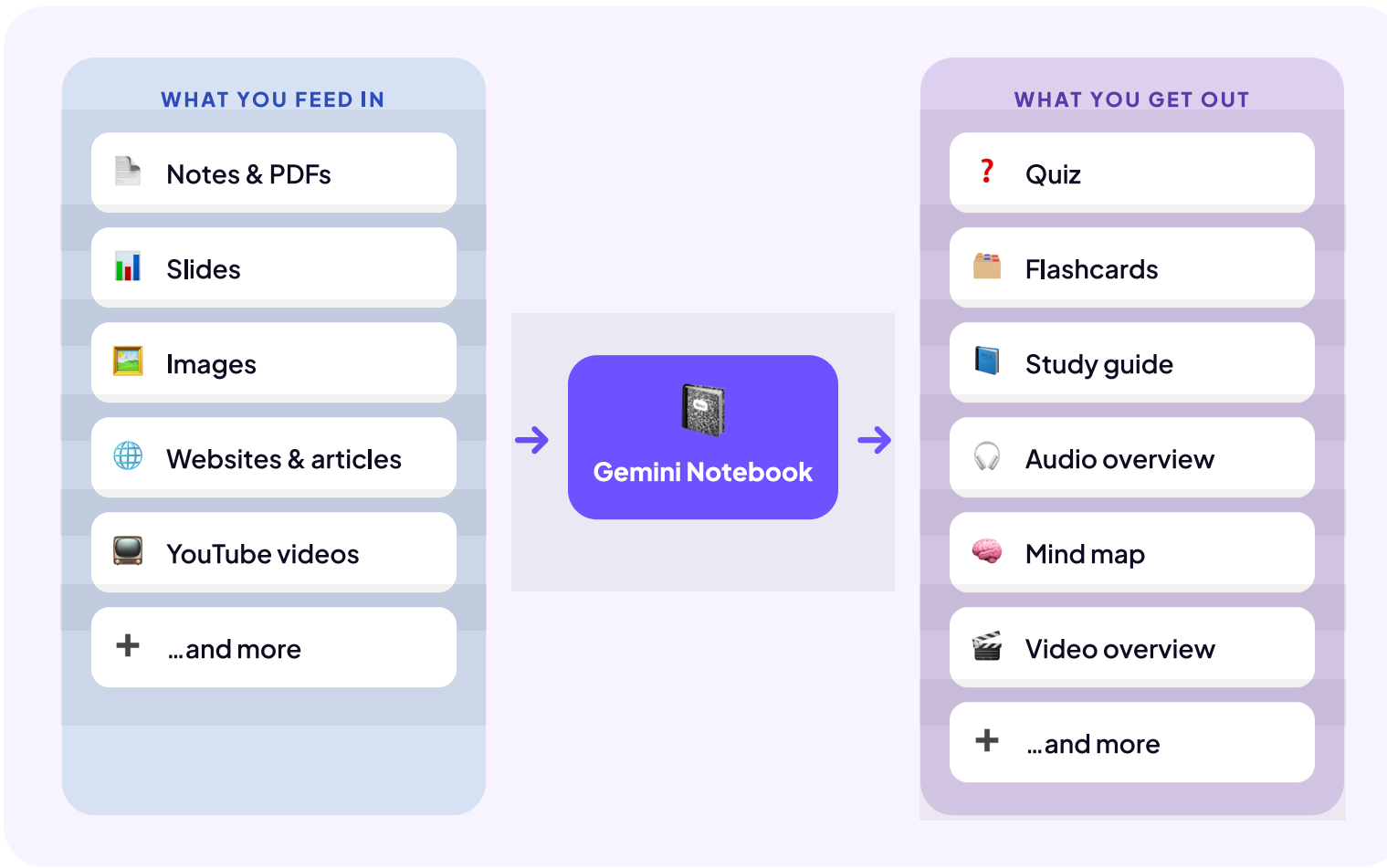
START HERE

Use Gemini Notebook as your main study tool. The clue is in the name: it was built for learning.

ChatGPT or Claude can do most of the same things, but it's more difficult because they weren't designed with the goal of learning.

HOW GEMINI NOTEBOOK WORKS

Feed it your own study materials, and it turns them into the kind of study tool that fits you best: audio to listen to, quizzes to test yourself, and more. You choose.



BEST PRACTICES

Here's how to use Gemini Notebook to its full potential.

1

One notebook per subject

Keep AP History and Chemistry in separate notebooks. Mixing sources confuses the model and gives you muddled study guides. Name each one clearly and treat it like its own binder.

2

Feed it a few angles, not one

For a big test, give it everything your teacher handed you: your class notes (so it keeps your teacher's wording), the slides, and any video or link the teacher shared. Gemini Notebook reads a video's transcript and blends it with your notes, so everything it gives back stays true to your class.

3

Quiz yourself blind

Re-reading notes feels productive, but that's recognition, not recall. Generate a quiz, close everything, and take it cold.

4

Review on the go

Have Gemini Notebook make an audio overview, then play it while you drive, walk, or work out. It turns dead time into another pass over the material.

5

Trace it back to learn it

When something doesn't fully click, click its numbered citation to jump straight to the spot in your materials it came from. Reading the original material is often what makes it land.



AI runs the drills. You build the skill.

Use AI to learn, not to skip the learning.

Feed it your materials. Trace it back.